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FIFTH SEMESTER **B.Tech. [ALL]**

END SEMESTER EXAMINATION ***Nov/Dec 2019***

IT 351 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time: 3:00 Hours ***Max. Marks: 50***

Note: Attempt total ***Five*** Questions and Question No. 1 is ***Compulsory***.
Assume suitable missing data, if any.

Question No. 1 [2x5=10]

[a] Describe the role of bias and activation function in an ANN.

[b] What's the difference between Type I and Type II error?

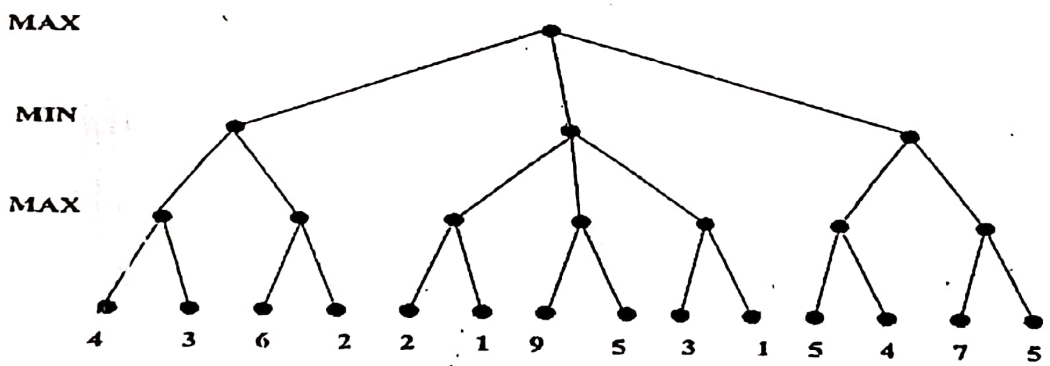
[c] Differentiate shallow and deep pruning in game trees with example.

[d] Explain any one method for encoding categorical inputs for neural networks.

[e] Discuss the experimental set-up and findings of Turing-test and Chinese room argument.

Question No. 2 [5x2=10]

[a] Apply alpha-beta pruning on the following tree.



[b] Suppose you are given the following positively labelled data points: {(3, 1), (3, -1), (6, 1), (6, -1)} and the following negatively labelled data points: {(1, 0), (0, 1), (0, -1), (-1, 0)}. Find the equation of the

separating hyperplane that accurately discriminates the two classes using Support Vector Machine.

Question No. 3 [5x2=10]

[a] Construct a truth table for $((p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s))$. Consider the proposition: "*If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area*". Translate this proposition into a logical expression.

[b] Consider the data points (x_1, x_2) with target t given in Table 1. The final output (y) is calculated as:

$$y = f(y_i) = \begin{cases} 1, & if\ y_i > 0.2 \\ 0, & if\ -0.2 \leq y_i \leq 0.2 \end{cases}$$

Assume initial weights and bias as '0' and learning rate as '1', find the weights and bias (till *Epoch 2*) required to perform the classification using the perceptron learning rule.

Table 1

x_1	x_2	t
1	1	1
1	0	1
0	1	1
0	0	-1

Question No. 4 [5x2=10]

[a] Four points A (1, 1), B (2, 1), C (4, 3) and D (5, 4) are lying on a 2-dimensional X-Y plane. Group them into 2 clusters by calculating Euclidean distance using a K-means clustering algorithm. Show all the intermediate centroids and clusters in each iteration.

[b] Discuss how CBOW (Continuous Bag of Words) and Skip-gram model can learn the weights for generating the word vector representation.

Question No. 5

[5x2=10]

[a] Consider the medical diagnosis data given in Table 2. Construct a decision tree on the following dataset using the ID3 algorithm.

Table 2 Medical Diagnosis Data

Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
Yes	Yes	Yes	Yes	Yes	Strep Throat
No	No	No	Yes	Yes	Allergy
Yes	Yes	No	Yes	No	Cold
Yes	No	Yes	No	No	Strep Throat
No	Yes	No	Yes	No	Cold
No	No	No	Yes	No	Allergy
No	No	Yes	No	No	Strep Throat
Yes	No	No	Yes	Yes	Allergy
No	Yes	No	Yes	Yes	Cold
Yes	No	No	Yes	Yes	Cold

[b] Briefly explain the following terminologies:

(i) K-fold cross-validation testing

(ii) Batch size

(iii) Bias-variance trade off

(iv) Multi-label classification

(v) Imbalanced dataset

Question No. 6

[5x2=10]

[a] Derive the regression coefficients of a linear regression model using the least-squares method.

[b] Discuss the importance of knowledge representation schemes. Compare and contrast semantic networks with Frames.